

INDEX

S.NO	DATE	TITLE
1.	21/1/25	Azure DevOps Environment Setup
2.	21/1/25	Azure DevOps Project Setup and User Story Management
3.	28/1/25	Setting Up Epics, Features and User Stories for Project Planning
4.	11/2/25	Sprint Planning
5.	18/2/25	Poker Estimation
6.	25/2/25	Designing Class and Sequence Diagrams for Project Architecture
7.	04/3/25	Designing Use-Case and Activity Diagrams for Project Architecture
8.	25/3/25	Testing – Test Plans and Test Cases
9.	15/4/25	CI/CD Pipelines in Azure
10.	22/4/25	GitHub: Project Structure & Naming Convention

EXP NO: 1 DATE:	AZURE DEVOPS ENVIRONMENT SETUP
----------------------------------	---------------------------------------

AIM:-

To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

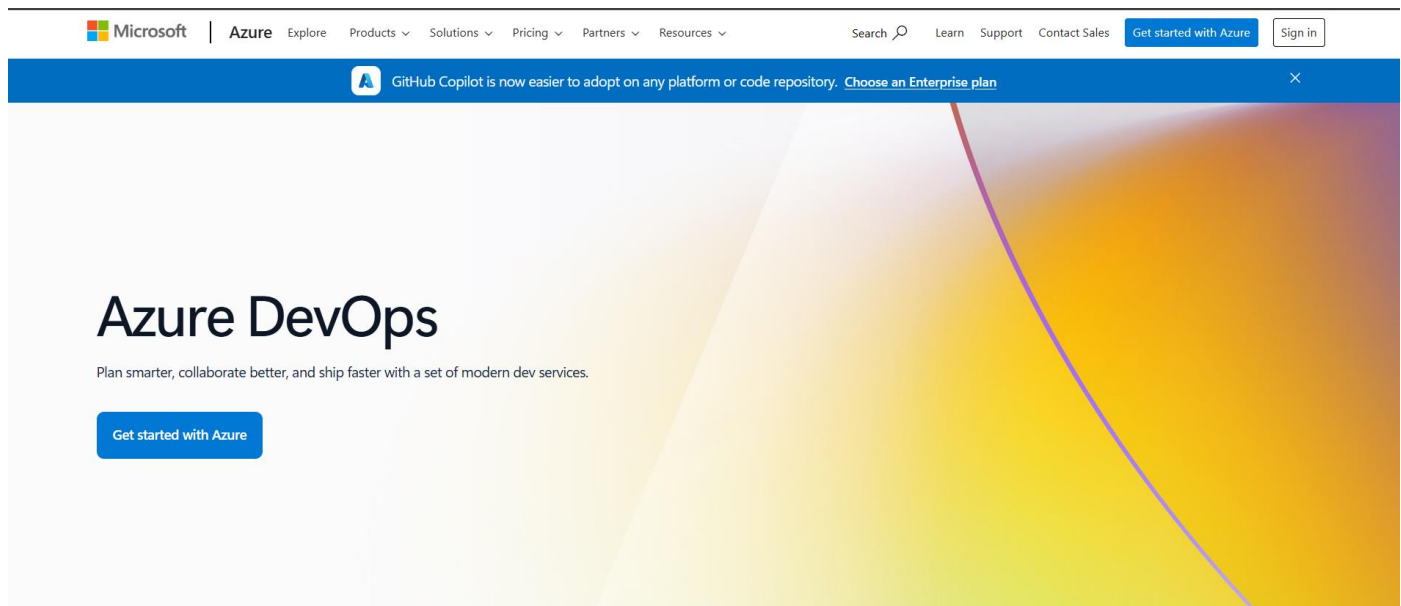
INSTALLATION:-

1. Open your web browser and go to the Azure website:

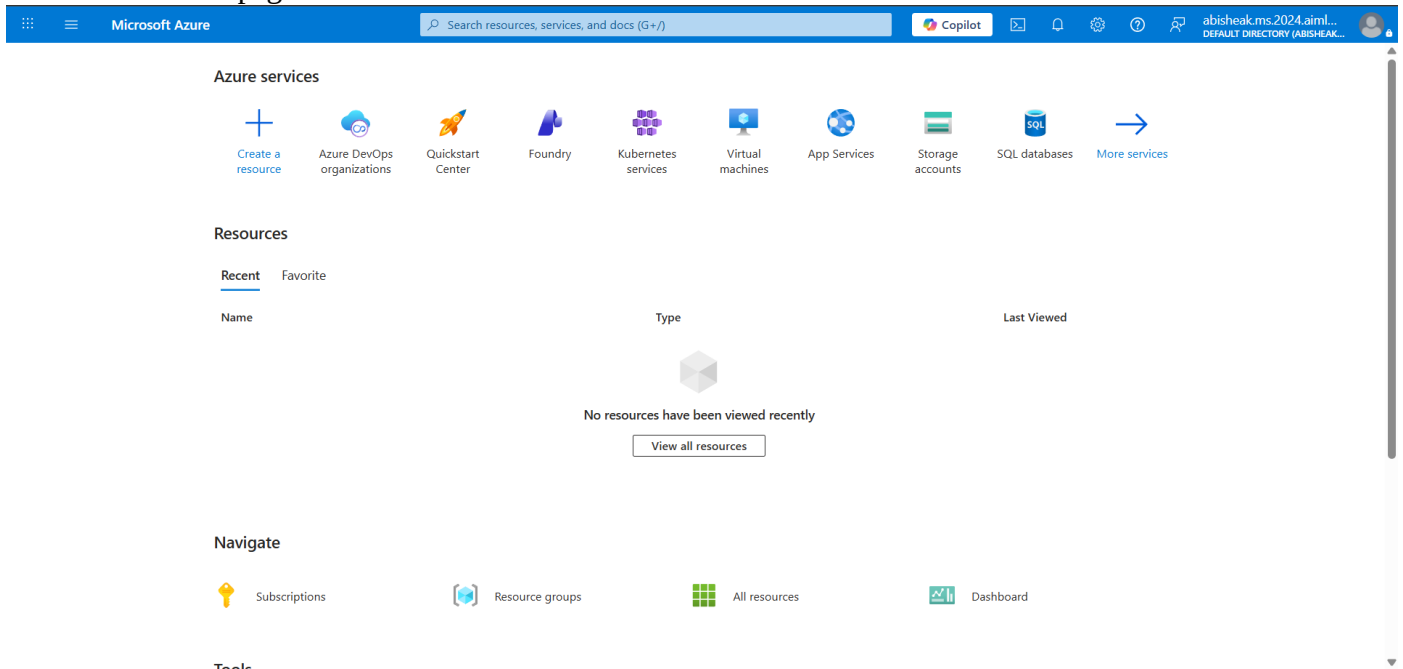
<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

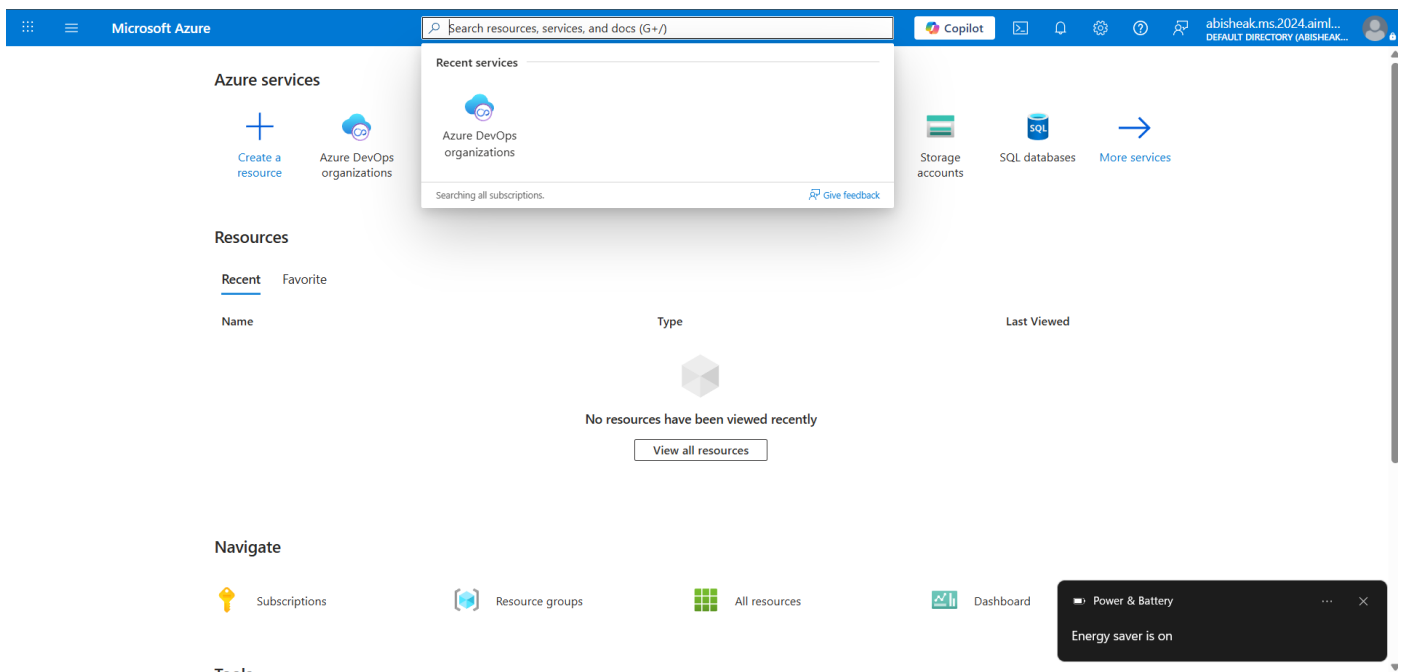
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



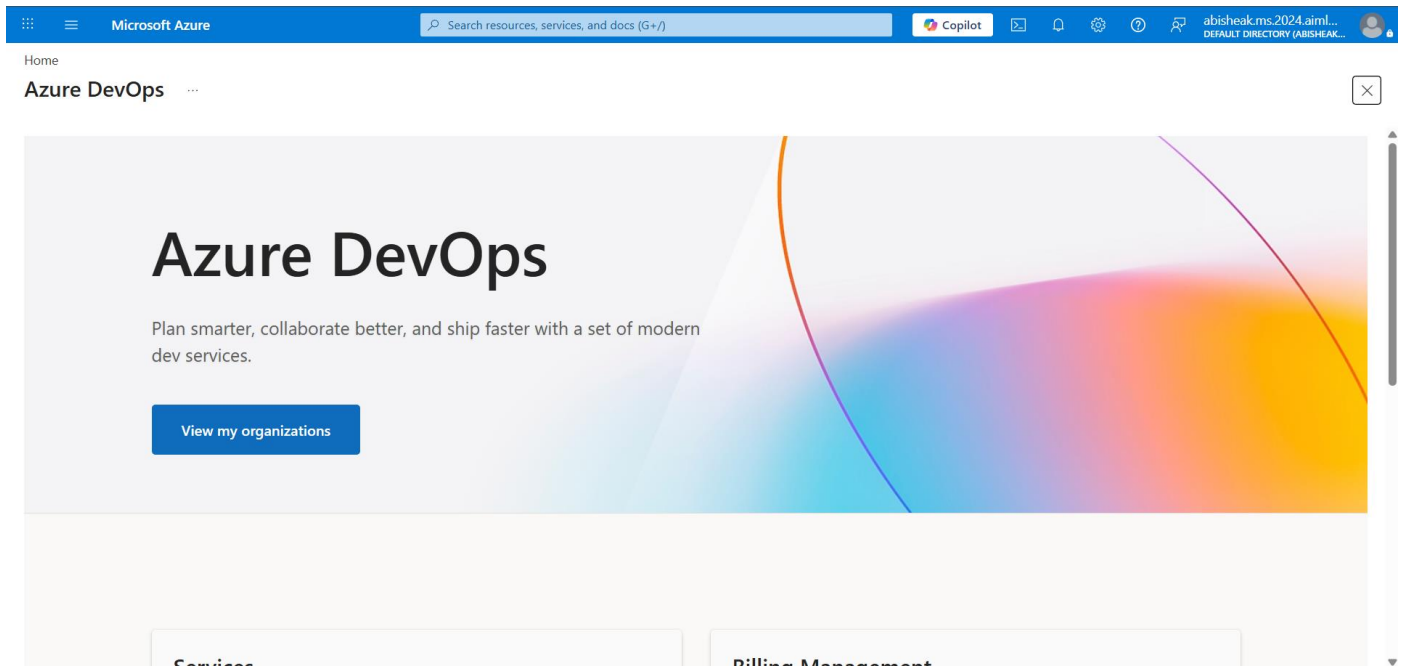
2. Azure home page



3. Open DevOps environment in the Azure platform by typing **Azure DevOps Organizations** in the search bar.



4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



RESULT:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

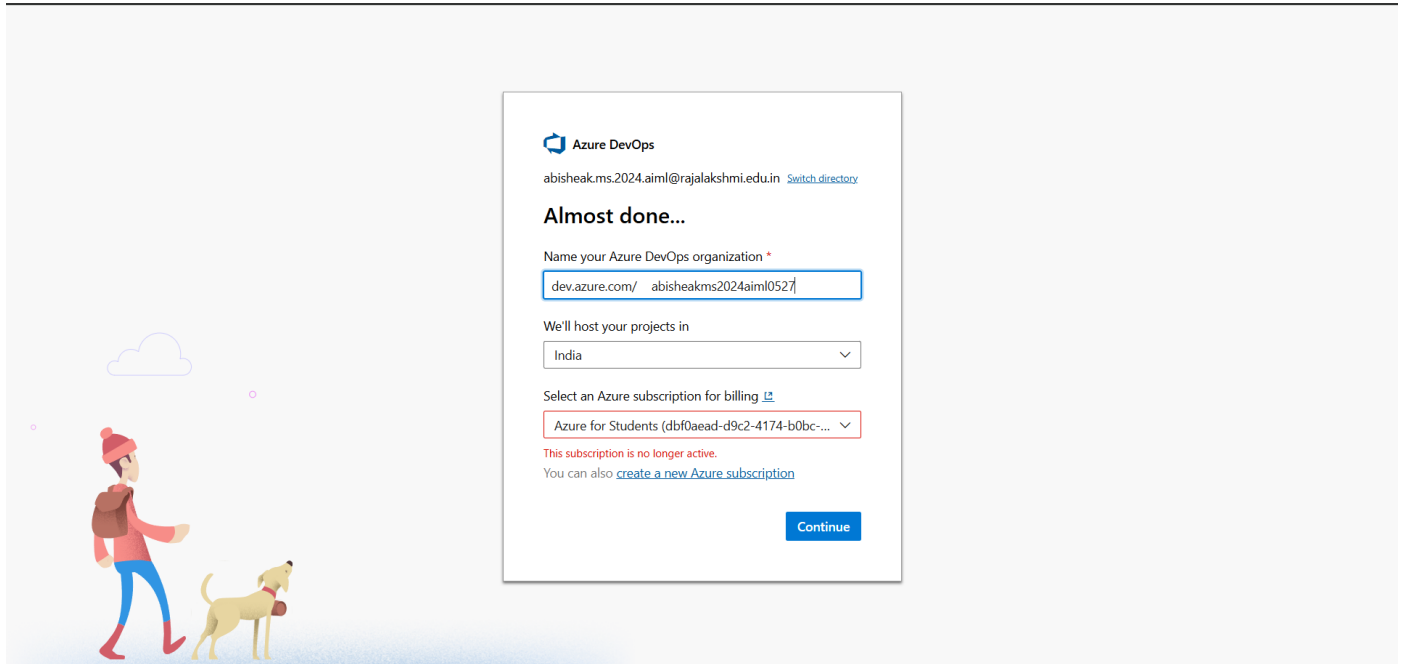
EXP NO: 2
DATE:

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

AIM:-

To set up an Azure DevOps project for efficient collaboration and agile work management.

1. Create An Azure Account



2. Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
Name: Choose a name for the project (e.g., **LMS**).
Description: Optionally, add a description to provide more context about the project.
Visibility: Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

Create new project



Project name *

Blog system

Description

Visibility



Private

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

▼ Advanced

Cancel

Create

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

Microsoft

AbisheakSign out

A

Abisheak

abisheak.ms.2024.ai@rajalakshmi.edu.in

Microsoft account

India

abisheak.ms.2024.ai@rajalakshmi.edu.in

Visual Studio Dev Essentials

Get everything you need to build and deploy your app on any platform.

Use your benefits

Azure DevOps Organizations

Create new organization

dev.azure.com/abisheakms2024aiml (Owner)

Projects

Blog Management System

online food

New project

Actions

Open in Visual Studio

dev.azure.com/maheendrans2024aiml (Member)

dev.azure.com/mishals2024aiml (Member)

4. Project dashboard

Azure DevOpsabisheakms2024aiml / Blog Management System / Overview / Summary

Search

PrivateInvite

BSBlog Management Syst...

Overview

Summary

Dashboards

Wiki

Boards

Repos

Pipelines

Test Plans

Artifacts

Project settings

BSBlog Management System

About this project

Help others to get on board!

Describe your project and make it easier for other people to understand it.

Add Project Description



Project stats

Period: Last 7 days

Boards

0 Work items created

0 Work items completed

Pipelines

0%

Builds succeeded

Members

4

A

MS

LP

MS

5. To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

The screenshot displays the Azure DevOps interface for the 'Blog Management System' project. The left-hand navigation menu is open, showing various project management tools. The 'Work Items' section is selected, displaying a list of work items. The table below represents the data shown in the screenshot.

ID	Title	Assigned To	State	Area Path	Tags	Comments	Activity Date
50	Implement pagination logic; Add next/prev buttons	Abisheak	Active	Blog Management System			3/30/2026 4:21:02 PM
49	Create categories; Filter blogs by category	Abisheak	Active	Blog Management System			3/30/2026 4:20:56 PM
48	Add search bar; Implement search logic; Display results	Abisheak	Active	Blog Management System			3/30/2026 4:19:23 PM
47	Create media gallery; Implement delete feature	Abisheak	Active	Blog Management System			3/30/2026 4:19:11 PM
46	Create upload UI; Validate file types; Store media	Abisheak	Active	Blog Management System			3/30/2026 4:19:03 PM
45	Add delete option; Verify permissions; Remove comment	Abisheak	Active	Blog Management System			3/30/2026 4:18:47 PM
44	Create comment input; Save comment; Display comment list	Abisheak	Active	Blog Management System			3/30/2026 4:18:31 PM
43	Implement logout function; Clear session data	Abisheak	Active	Blog Management System			3/30/2026 4:18:24 PM
42	Create login UI; Authenticate credentials; Maintain session	Abisheak	Active	Blog Management System			3/30/2026 4:18:17 PM
41	Create registration UI; Validate user input; Store user data securely	Abisheak	Active	Blog Management System			3/30/2026 4:18:08 PM
40	Add publish button; Update post status; Display published date	Abisheak	Active	Blog Management System			3/30/2026 4:18:02 PM
39	Fetch existing post data; Update UI form; Implement update logic; Save	Abisheak	Active	Blog Management System			3/30/2026 4:17:55 PM
38	Design blog form UI; Implement create post API; Validate input fields;	Abisheak	Active	Blog Management System			3/30/2026 4:11:39 PM
37	Blog Pagination	Abisheak	Active	Blog Management System			3/30/2026 4:10:42 PM
36	Categorized Archives	Abisheak	Active	Blog Management System			3/30/2026 4:10:37 PM
35	Search Blog Content	Abisheak	Active	Blog Management System			3/30/2026 4:10:29 PM
34	Manage Media	Abisheak	Active	Blog Management System			3/30/2026 4:10:23 PM
33	Upload Images & Videos	Abisheak	Active	Blog Management System			3/30/2026 4:10:18 PM

Result

Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3
DATE:

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

AIM:-

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface for a team named 'E-commerce Product Uploader Team'. The left sidebar contains navigation options: Overview, Boards, Work Items, Backlogs (selected), Sprints, Queries, Delivery Plans, Analytics views, Planning Poker, Repos, Pipelines, Test Plans, and Project settings. The main area displays the 'Backlog' view with a table of work items. The table has columns for Order, Work Item Type, Title, State, Effort, Business Value, and Tags. The work items are organized into three epics: 1. Product Search & Pagination, 2. Bulk Product Upload & Validation, and 3. Product Management (CRUD Operations). Each epic contains several features and user stories, all marked as 'Active'.

Order	Work Item Type	Title	State	Effort	Business Value	Tags
1	Epic	Product Search & Pagination	Active		Business	
	Feature	Product Search	Active		Business	
	User Story	As a store owner, I want to search products by name s...	Active		Business	
	User Story	As a store owner, I want to search products by price s...	Active		Business	
	User Story	As a store owner, I want to view search results clearly ...	Active		Business	
	Feature	Pagination for Product Listing	Active		Business	
	User Story	As a store owner, I want to see only 10 products per p...	Active		Business	
	User Story	As a store owner, I want pagination controls so that I c...	Active		Business	
	User Story	As a store owner, I want pagination to work with sear...	Active		Business	
2	Epic	Bulk Product Upload & Validation	Active		Business	
	Feature	Bulk Upload via CSV/Excel	Active		Business	
	Feature	Image & Content Validation	Active		Business	
3	Epic	Product Management (CRUD Operations)	Active		Business	
	Feature	Manage Product Details	Active		Business	

1.Fill in Epics

The screenshot shows the details of a work item titled 'User Authentication' (ID 7) under the epic 'User Authentication'. The work item is created by 'Abisheak' and is currently 'Active'. It is associated with the 'Blog Management System' area and the 'Blog Management System/Sprint 3' iteration. The 'Description' field is empty. The 'Discussion' section shows a comment box with a placeholder text: 'Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.' The 'Planning' section shows 'Priority' 2 and 'Risk' 1. The 'Deployment' section has a note: 'To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting.' The 'Development' section has a note: 'Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.' The 'Related Work' section shows a list of child items: '20 User Login', '21 User Logout', and '8 User Registration', all updated on Mar 30 and marked as 'Active'. The 'Planning Poker' section shows a 'Voted 0' button.

2116231501146

CS23432

2.Fill in Features

FEATURE 8

8 User Registration

A Abisheak

0 Comments Add Tag

State Active

Area Blog Management System

Reason Implementation started Iteration Blog Management System\Sprint 3

Description

B I U [List Bulleted] [List Numbered] Undo Redo Bold Italic Underline Link Image Video Embed Code Preview Full Screen Exit Help

[switch to Markdown editor](#)

Discussion

A Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

[switch to Markdown editor](#)

Planning

Priority 2

Risk

Effort

Business Value

Time Criticality

Start Date Select a date...

Target Date Select a date...

Classification

Value area Business

Voted: 0

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link
Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link ▾

Parent

- User Authentication Updated Mar 30 ● Active

Child

- Register a new User Updated Mar 30 ● Active

3.Fill in User Story Details

User Story 9

9 Register a new User

Abisheak

0 Comments Add Tag

Status Active

Area Blog Management System

Reason Implementation started Iteration Blog Management System/Sprint 3

Description

[switch to Markdown editor](#)

Acceptance Criteria

Click to add Acceptance Criteria.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

[switch to Markdown editor](#)

Planning

Story Points

Priority 2

Risk

Classification

Value area Business

Planning Poker

Voted: 0

0 1 2 3 5 8 13

Write your comment here

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add Link

Parent

- 8 User Registration Updated Mar 30 ● Active

Child

- 41 Create registration UI; Validate user input; Store user dat... Updated Mar 30 ● Active

RESULT:-

Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4
DATE:

SPRINT PLANNING

AIM:-

To assign user story to specific sprint for the **E-commerce Product Uploader** Project.

Sprint Planning:-

Sprint 1

The screenshot displays the Azure DevOps Sprints interface for the 'Blog Management System Team'. The board is set for 'Sprint 3' (March 4 - March 18, 11 work days) and shows a Kanban-style workflow with columns: New, Active, Resolved, and Closed. The 'Active' column is currently selected. The board contains several user stories, each with a title, ID, and assignee (Abhishek). The stories are organized into rows, with some collapsed. The left sidebar shows the project structure, including Overview, Boards, Work Items, Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Planning Poker, Repos, Pipelines, Test Plans, and Artifacts. The top navigation bar includes 'Azure DevOps', the user profile, and search options.

Item ID	Title	Status	Assignee
5	Create a Blog Post	Active	Abhishek
9	Register a new User	Active	Abhishek
25	Publish Blog	Active	Abhishek
26	Edit Existing Blog	Active	Abhishek
29	User Login	Active	-
38	Design blog form UI; Implement create post API; Validate input fields; Save post to database	Active	Abhishek
41	Create registration UI; Validate user input; Store user data securely	Active	Abhishek
40	Add publish button; Update post status; Display published date	Active	Abhishek
39	Fetch existing post data; Update UI form; Implement update logic; Save changes	Active	Abhishek
42	Create login UI; Authenticate credentials; Maintain session	Active	-

Sprint 2

Azure DevOps | abishakeem2024amr / Blog Management System / Boards / Sprints

Blog Management System Team

Taskboard Backlog Capacity Analytics

Sprint 1

March 4 - March 18
11 work days

Order	Title	State	Assigned To	Rem...
1	> Create a Blog Post	Active	Abishake	
2	> Register a new User	Active	Abishake	
3	> Publish Blog	Active	Abishake	
4	> Edit Existing Blog	Active	Abishake	
5	> User Login	Active	Abishake	
6	> User Logout	Active	Abishake	
7	> Comment on Blog	Active	Abishake	
8	> Comment on Blog	Active	Abishake	
9	> Upload Images & Videos	Active	Abishake	
10	> Manage Media	Active	Abishake	
11	> Search Blog Content	Active	Abishake	
12	> Categorized Archives	Active	Abishake	
13	> Blog Pagination	Active	Abishake	

RESULT:-

The Sprints are created for the **E-commerce Product Uploader**.

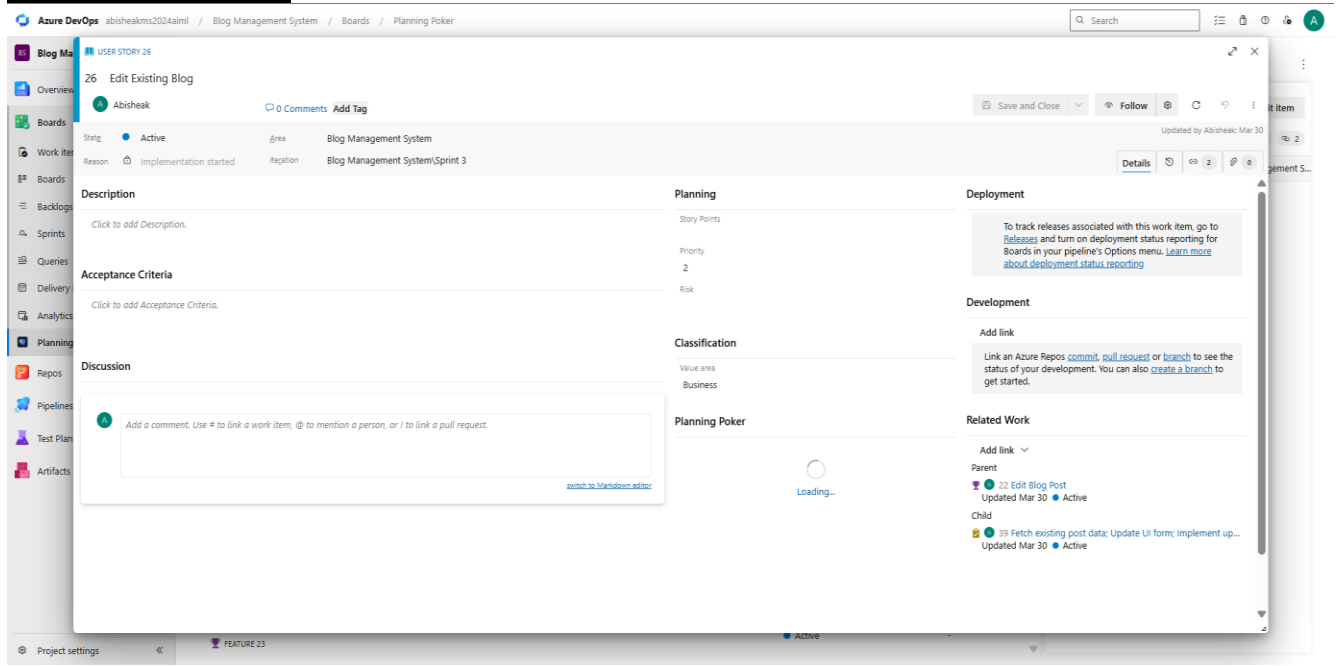
EXP NO: 5
DATE:

POKER ESTIMATION

AIM:-

Create Poker Estimation for the user stories - **E-commerce Product Uploader**.

Poker Estimation



The screenshot displays the Azure DevOps interface for a user story titled "26 Edit Existing Blog". The interface is divided into several sections:

- Description:** A section for adding a description, currently empty.
- Acceptance Criteria:** A section for adding acceptance criteria, currently empty.
- Discussion:** A section for adding comments. A comment box is visible with the text: "Add a comment. Use # to link a work item, @ to mention a person, or / to link a pull request."
- Planning:** A section for planning the work item. It includes fields for Story Points, Priority (set to 2), and Risk.
- Classification:** A section for classifying the work item. It includes fields for Value area (set to Business) and Business.
- Deployment:** A section for deployment. It includes a link to track releases associated with the work item.
- Related Work:** A section for related work. It includes a list of related items, such as "22 Edit Blog Post" and "39 Fetch existing post data; Update UI form; Implement up...".

Azure DevOps

abisheakms2024aiml

Blog Management System

/ Boards

/ Planning Poker

Q Search

Blog Management Syst...

+

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Planning Poker

Repos

Pipelines

Test Plans

Artifacts

Project settings

← Poker game, Mar 31, 2026

Current item: Implement pagination logic; Add next/prev buttons

Participants: 1 Voted: 1 Share game Become spectator

2

Type here Save & next Edit vote Reset

> Reference Items

Game scope

Work item	State	Estimates
Implement pagination logic; Add next/prev buttons TASK 50	Active	-
Create categories; Filter blogs by category TASK 49	Active	-
Add search bar; Implement search logic; Display results TASK 48	Active	-
Create media gallery; Implement delete feature TASK 47	Active	-
Create upload UI; Validate file types; Store media TASK 46	Active	-
Add delete option; Verify permissions; Remove comment TASK 45	Active	-
Create comment input; Save comment; Display comment list TASK 44	Active	-
Implement logout function; Clear session data TASK 43	Active	-
Create login UI; Authenticate credentials; Maintain session		

Next

TASK 50

Implement pagination logic; Add next/prev buttons

Edit item

Abisheak

0 0 1

State Active Iteration Blog Management S...

Description is empty

← Poker game, Mar 31, 2026

Current item: User Login

Participants: 1 Voted: 0 Share game Become spectator

Reveal

0 1 2 3 5 8 13

> Reference Items

Game scope

Work item	State	Estimates
Implement pagination logic; Add next/prev buttons TASK 50	Active	-
Create categories; Filter blogs by category TASK 49	Active	-
Add search bar; Implement search logic; Display results TASK 48	Active	-
Create media gallery; Implement delete feature TASK 47	Active	-
Create upload UI; Validate file types; Store media TASK 46	Active	-
Add delete option; Verify permissions; Remove comment TASK 45	Active	-

Next

USER STORY 29

User Login

Edit item

Abisheak

0 0 2

State Active Iteration Blog Management S...

Description is empty

RESULT:-

The Estimation/Story Points is created for the project using Poker Estimation.

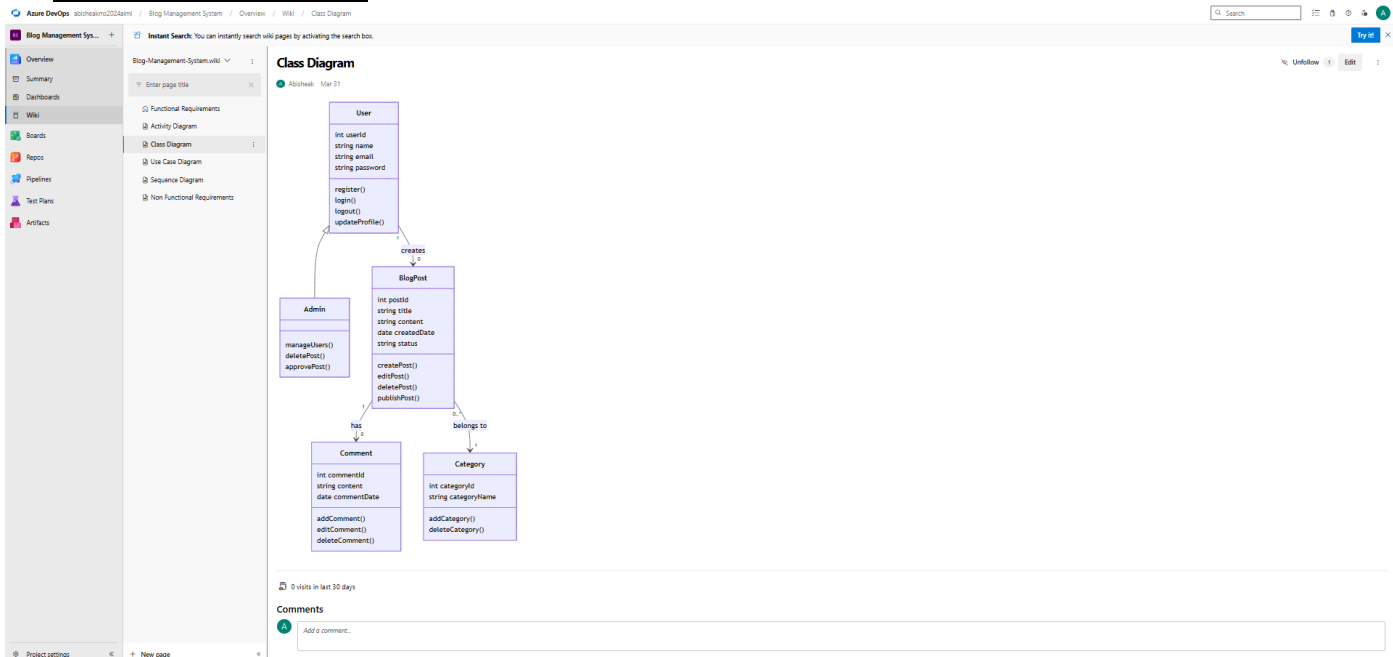
EXP NO: 6
DATE:

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

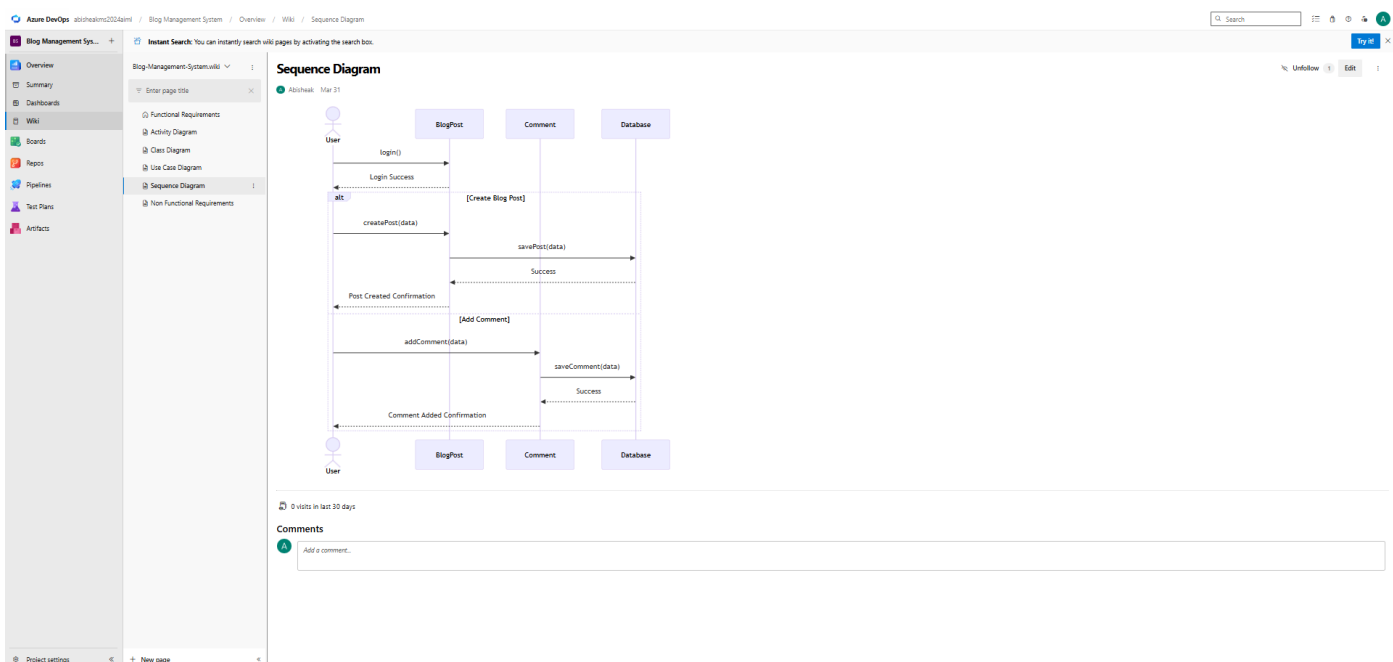
AIM:-

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. CLASS DIAGRAM:-



6B. SEQUENCE DIAGRAM:-



Result:

The Class Diagram and Sequence Diagram is designed Successfully for given project.

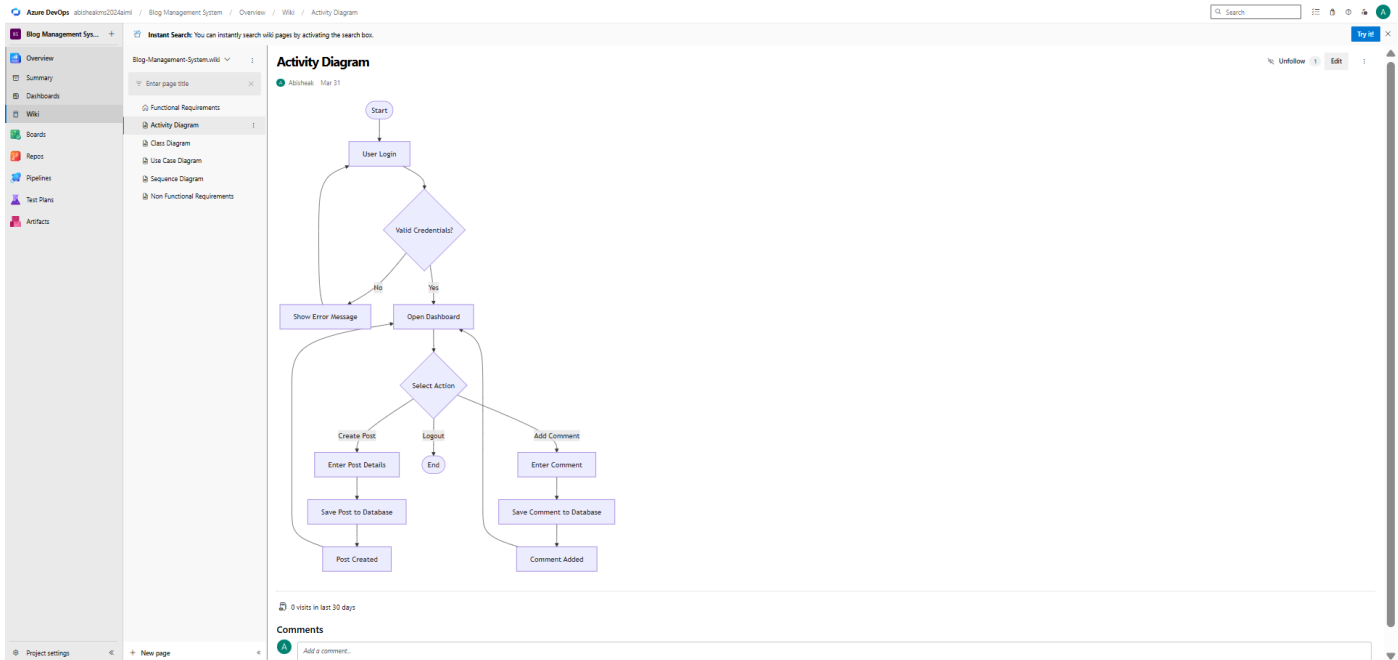
EXP NO: 7
DATE:

DESIGNING USE CASE DIAGRAMS AND ACTIVITY DIAGRAM FOR PROJECT ARCHITECTURE

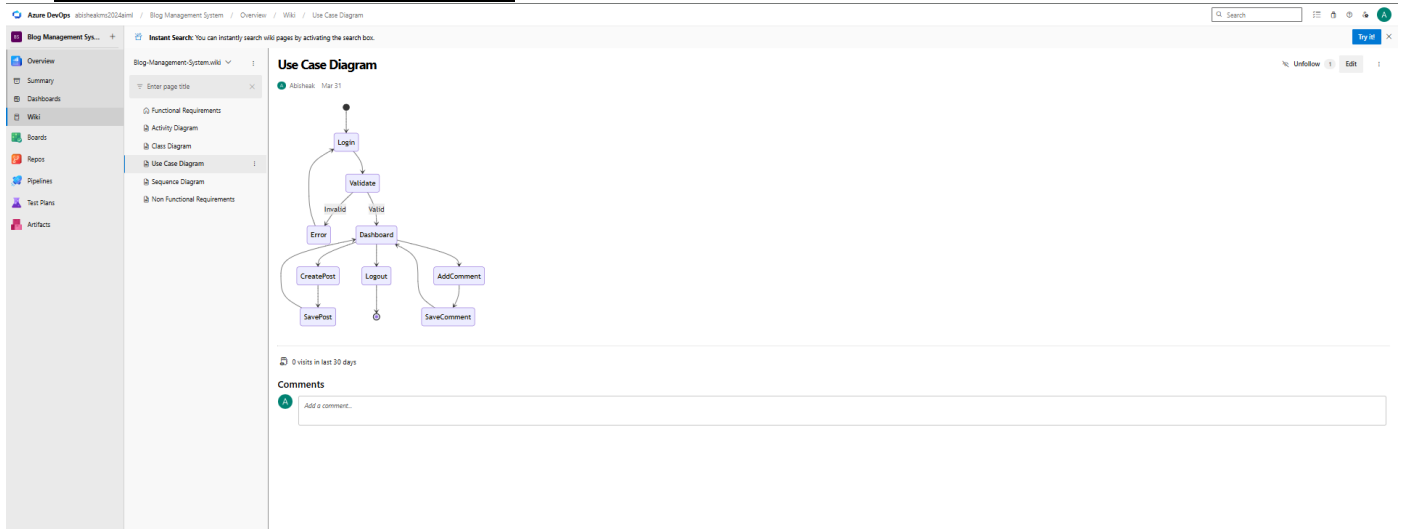
AIM:-

To Design an activity Diagram and use case Diagram for the given Project.

7A. Activity Diagram:-



7B. USE CASE DIAGRAM:-



RESULT:-

Thus activity and use case diagram has been designed successfully for **E-commerce Product Uploader**.

EXP NO: 8 DATE:	TESTING – TEST PLANS AND TEST CASES
----------------------------------	--

AIM:-

To create Test Plan and Test Cases and write two test cases for at least five user stories (Happy Path & Error Path) for the E-commerce Product Upload System in the Azure DevOps platform.

Test Planning and Test Case
Test Case Design Procedure**1. Understand Core Features of the Application**

- Admin Login & Authentication
- Upload Product (CSV/Excel)
- Product Data Validation
- Product Storage in Database
- View Upload Status / Reports

2. Define User Interactions

- Each test case represents real user actions
 - Admin logs in
 - Uploads product file
 - Views upload result

3. Design Happy Path Test Cases

- Ensure system works correctly under valid conditions
- Example:
 - Admin uploads valid file → Data saved successfully → Success message displayed

4. Design Error Path Test Cases

- Check system behavior for invalid inputs
- Example:
 - Upload invalid file → Error message shown
 - Missing fields → Validation fails

5. Break Down Steps and Expected Results

- Each test case includes:
 - Step-by-step actions
 - Expected output
 - Helps manual testing & automation

6. Use Clear Naming and IDs

- Example:

- TC01 – Admin Login Success
- TC02 – Invalid Login
- TC10 – Product Upload Failure

7. Separate Test Suites

- Login & Authentication
- Product Upload
- Validation
- Database Storage
- Report / Status Viewing

8. Prioritize and Review

- High priority:

- Login
- Product Upload
- Validation
 - Reviewed for correctness and completeness

1. New test plan

NEW TEST CASE *

TC-1

Barath D 0 Comments Add Tag Save and Close

State Design Area E-commerce Product Uploader Reason New Iteration E-commerce Product Uploader\Sprint 1

Steps

Click or type here to add a step

Recent test results

Run manual tests with web runner. [Learn how to get started](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also create a branch to get started.

Related Work

Add link

2.Test case:

The screenshot displays a test management interface. On the left is a sidebar with navigation links: Overview, Boards, Repos, Pipelines, Test Plans, Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. The main area is divided into two panels. The left panel, titled 'Test Suites', shows a summary for 'Feb 2 - Feb 16' with a 'Pass' status, '100% run, 100% passed', and a 'View report' link. Below this is a filter bar 'Filter suites by name' and a list with one item, 'login (1)'. The right panel, titled 'Test Cases (5 items)', has tabs for 'Define', 'Execute', and 'Chart'. It contains a table with columns: Title, Order, Test Case Id, Assigned To, and State. A 'New Test Case' button is in the top right of the table.

Title	Order	Test Case Id	Assigned To	State
TC-01 Login with valid credentials	1	51	Barath D	Design
TC-02 Login with invalid password	2	52	Barath D	Design
TC-03 Login with invalid username	3	54	Barath D	Design
TC-04 Remember me functionality	4	57	Barath D	Design
TC-05 Logout functionality	5	58	Barath D	Design

2. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

E-commerce Product Uploader– Test Plans

USER STORIES:-

- As an admin, I want to log in securely so I can access the system. (ID: 201)
- As an admin, I want to upload product files (CSV/Excel) to add products. (ID: 202)
- As an admin, I want the system to validate product data to ensure correctness. (ID: 203)
- As an admin, I want product data to be stored in the database. (ID: 204)
- As an admin, I want to view upload status and reports. (ID: 205)

Test Suites:-

◆ **User Story 1: Admin Login**

TC01 – Happy Path

- Enter valid username & password
 - Click login
 - Expected: Login successful, dashboard displayed
- TC02 – Error Path
- Enter wrong credentials
 - Expected: Error message "Invalid login"

◆ **User Story 2: Upload Product File**

TC03 – Happy Path

- Upload valid CSV file
 - Expected: File uploaded successfully
- TC04 – Error Path
- Upload unsupported file format
 - Expected: Error "Invalid file format"

◆ **User Story 3: Validate Product Data**

TC05 – Happy Path

- File contains valid product details
 - Expected: Validation successful
- TC06 – Error Path
- Missing product fields
 - Expected: Validation error shown

◆ **User Story 4: Save Product Data**

TC07 – Happy Path

- Valid data sent to database
 - Expected: Data stored successfully
- TC08 – Error Path
- Database failure
 - Expected: Save failed message

◆ **User Story 5: View Upload Status**

TC09 – Happy Path

- View uploaded products
Expected: Correct data displayed
TC10 – Error Path
- No data available
Expected: "No records found" message

1. Test Cases

TEST CASE 51

51 TC-01 Login with valid credentials

0 Comments Add Tag

Save and Close Follow

Updated by Barath D: 23 Mar

State Design Area E-commerce Product Uploader Reason New Regression E-commerce Product Uploader\Sprint 1

Steps Summary Associated Automation

Steps

1. Open login page

User is redirected to dashboard

2. Enter valid username

User is redirected to dashboard

3. Enter valid password

User is redirected to dashboard

4. Click Login

User is redirected to dashboard

Click or type here to add a step

Recent test results

Passed by with Windows 10

Completed 12:27 PM, 4/6/2026

Failed by with Windows 10

Completed 12:13 PM, 4/6/2026

Passed by with Windows 10

Completed 12:13 PM, 4/6/2026

Failed by with Windows 10

Completed 7:09 AM, 3/23/2026

View all test result history

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

Add link

2.

Define Execute Chart

Test Cases (5 items)

New Test Case

☐ Title	Order	Test Case Id	Assigned To	State
☐ TC-01 Login with valid credentials	1	51	Barath D	Design
☐ TC-02 Login with invalid password	2	52	Barath D	Design
☐ TC-03 Login with invalid username	3	54	Barath D	Design
☐ TC-04 Remember me functionality	4	57	Barath D	Design
☐ TC-05 Logout functionality	5	58	Barath D	Design

0 Comments [Add Tag](#)

 Save and Close Follow

Updated by Barath D: 6 Apr

Area

Steps

Summary

Associated Automation

0

0




 |
 


 |
 

 |
 B
I
U

[Click or type here to add a step](#)

 Loading

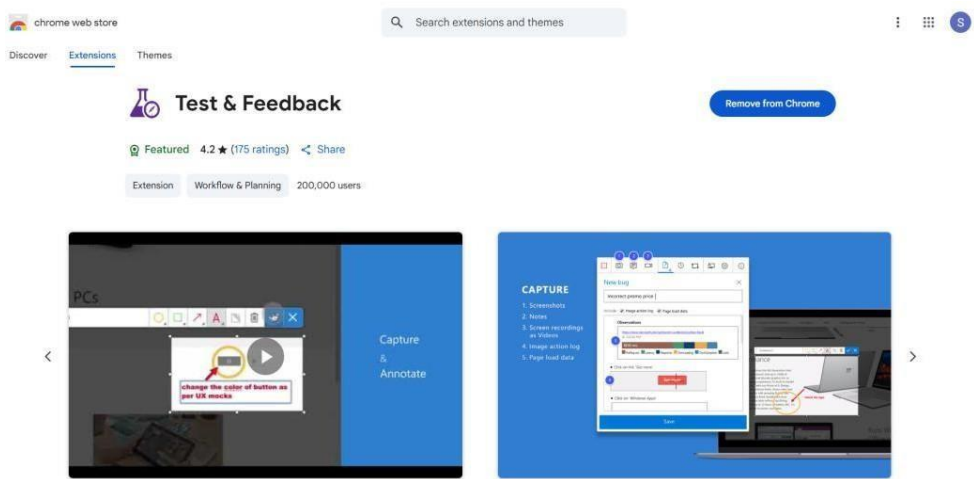
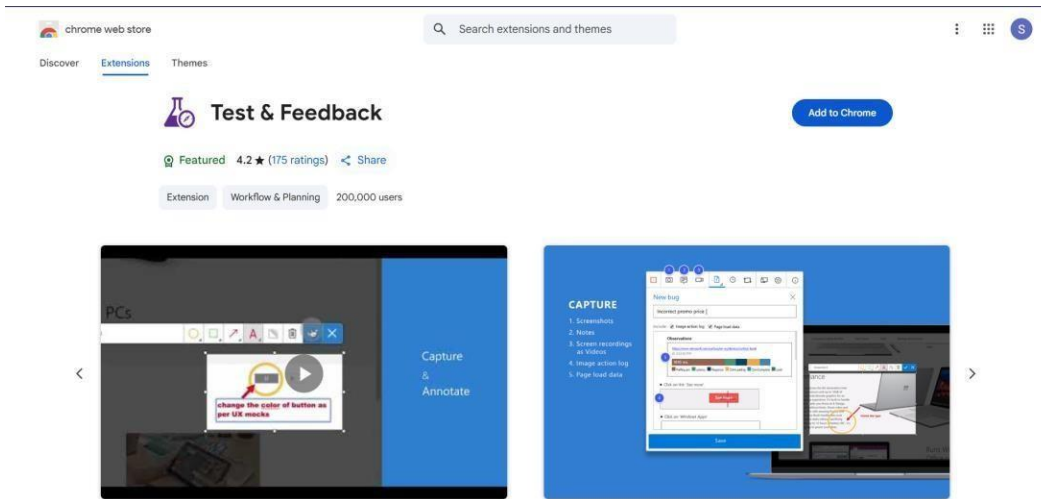
To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more](#) **Loading...** [deployment status reporting](#)

Add link

 Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Add link

1. Installation of test



2. Running the test cases

The screenshot displays the Test Suite interface. On the left, a sidebar contains navigation links: Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. At the bottom of the sidebar is Project settings. The main area shows the Test Suites section for the 'login (5)' suite, indicating 100% run and 100% passed. The Test Points table lists five items, with the first item selected and a context menu open.

Title	Outcome	Order	Test Case Id	State
TC-01 Login with valid credentials	Passed	1	51	Design
TC-02 Login with invalid password	Passed	2	52	Design
TC-03 Login with invalid username	Passed	3	54	Design
TC-04 Remember me functionality			57	Design
TC-05 Logout functionality			58	Design

The context menu for the selected item includes the following options:

- View execution history
- Mark Outcome
- Run
- Reset test to active
- Edit test case
- Assign tester
- View test result

The 'Run' option is highlighted, and a sub-menu is visible with the following options:

- Run for web application
- Run for desktop application
- Run with options

 Save and close |  Create bug |  |  |  | ...

51: TC-01 Login with valid credentials  |  ▼

1.  Open login page  

EXPECTED RESULT
User is redirected to dashboard

2. Enter valid username  

EXPECTED RESULT
User is redirected to dashboard

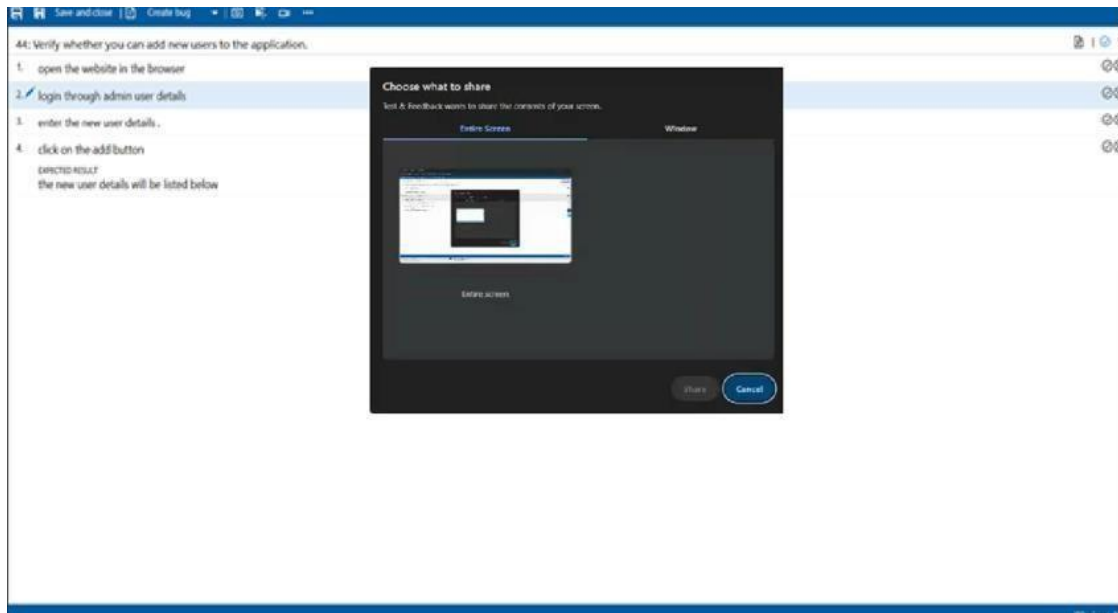
3. Enter valid password  

EXPECTED RESULT
User is redirected to dashboard

4. Click Login  

EXPECTED RESULT
User is redirected to dashboard

2. Recording the test case



3. Creating the bug

dev.azure.com/barathd2024aiml/E-commerce%20Product%20Uploader/_testExecution/Index

NEW BUG *

Invalid Login

No one selected 0 Comments Add Tag Save and Close

State: New Area: E-commerce Product Uploader Reason: New Iteration: E-commerce Product Uploader\Sprint 1

Details 2 0

Repro Steps

5/3/2026 7:56 AM Bug filed on "TC-01 Login with valid credentials"

Step no.	Result	Title
1.	None	Open login page
		<u>Expected Result</u>
		User is redirected to dashboard
2.	None	Enter valid username
		<u>Expected Result</u>
		User is redirected to dashboard
3.	None	Enter valid password
		<u>Expected Result</u>
		User is redirected to dashboard

Planning

Resolved Reason

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also create a branch to get started.

Related Work

Add link

Add an existing work item as a parent

Tested By

51 TC-01 Login with valid credentials
Updated 23 Mar Design

4. Test Results

The screenshot displays a test management interface. On the left is a sidebar with navigation links: Overview, Boards, Repos, Pipelines, Test Plans, Test plans, and Progress report. The main area is titled 'Test Suites' and shows a filter 'Filter suites by name' with a search bar. Below the filter, a list of test suites is shown, with 'login (3)' selected. The right panel shows the 'Test Points (5 items)' table, which lists five test cases, all of which have passed.

Feb 2 - Feb 16 **Pass**
100% run, 100% passed. [View report](#)

Test Suites

Filter suites by name

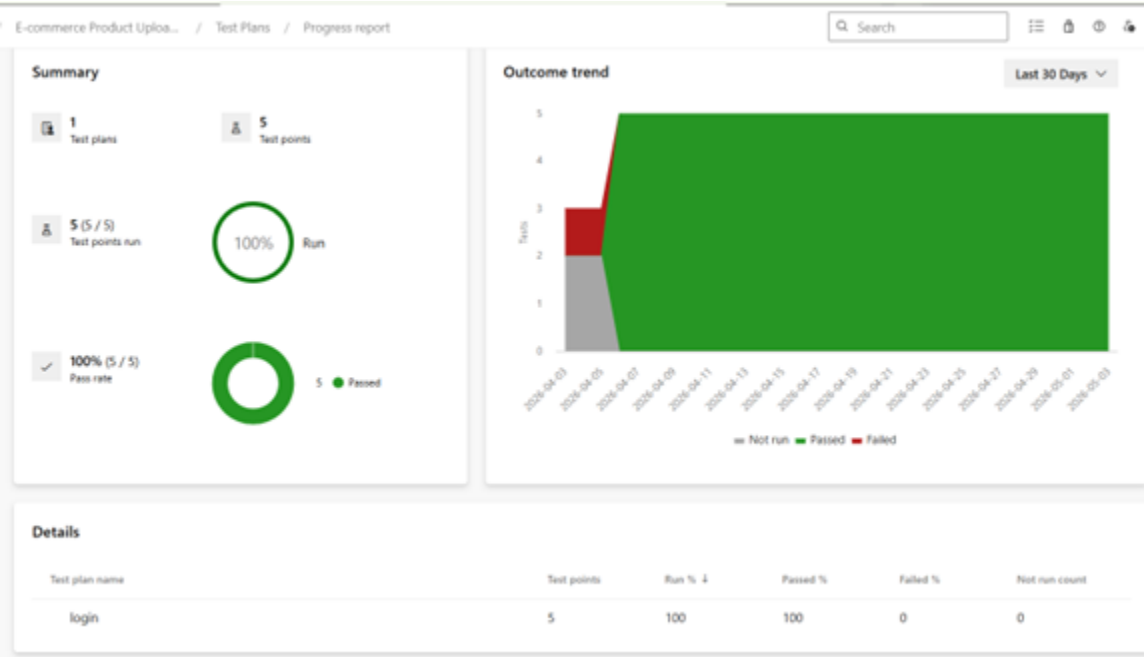
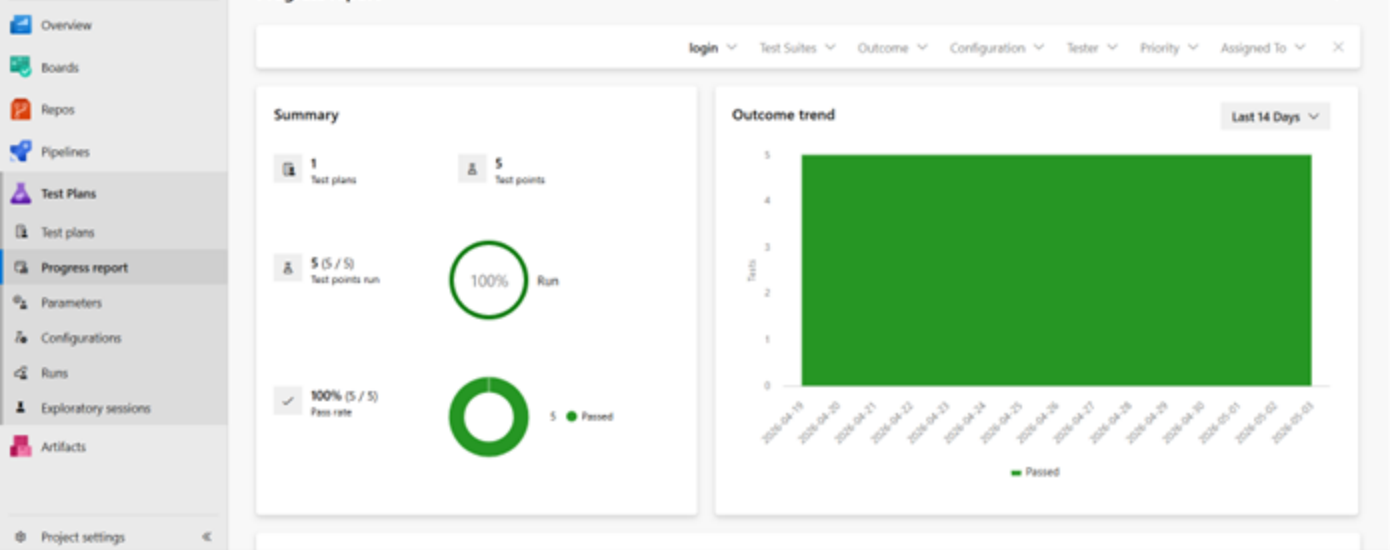
login (3)

Define **Execute** Chart

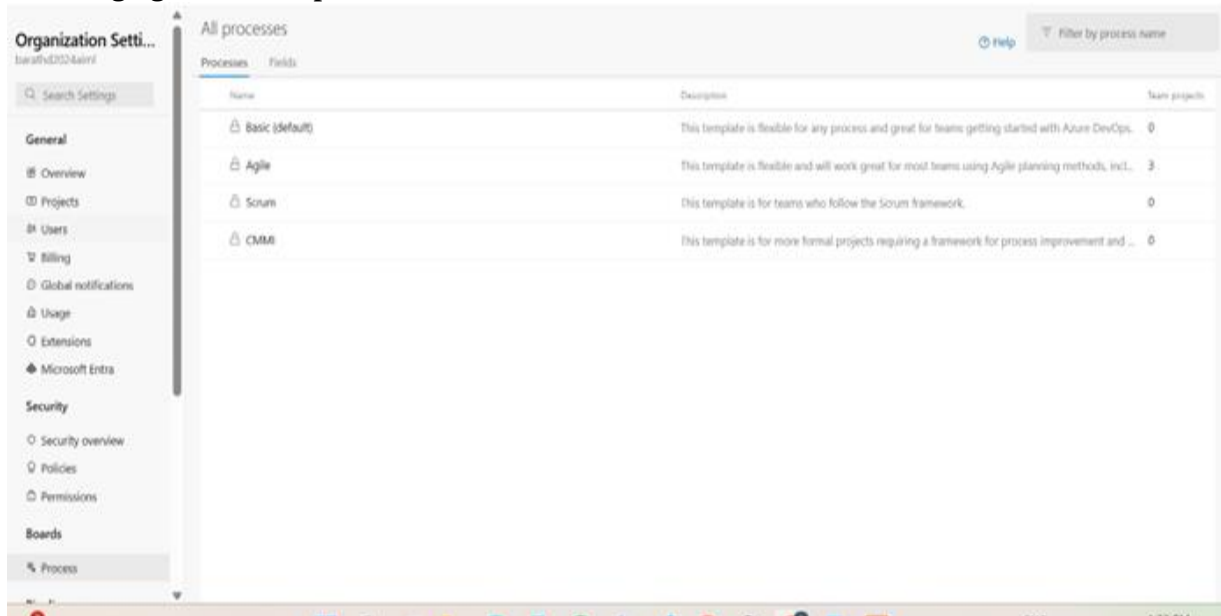
Test Points (5 items)

Title	Outcome	Order	Test Case Id	State
☒ TC-01 Login with valid credentials	Passed	1	51	Design
☐ TC-02 Login with invalid password	Passed	2	52	Design
☐ TC-03 Login with invalid username	Passed	3	54	Design
☐ TC-04 Remember me functionality	Passed	4	57	Design
☐ TC-05 Logout functionality	Passed	5	58	Design

Progress report



4. Changing the test template



The screenshot shows the 'All processes' page in Azure DevOps. The left sidebar contains the 'Organization Settings' menu with sections for General, Security, and Boards. The 'Processes' tab is selected, displaying a table of available templates. The table has columns for Name, Description, and Team projects.

Name	Description	Team projects
Basic (default)	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, incl...	3
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and ...	0

Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9 DATE:	CI/CD PIPELINES IN AZURE
----------------------------------	---------------------------------

AIM:

To implement a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automating the build, testing, and deployment process of the E-commerce product uploader, ensuring faster delivery and improved software quality.

PROCEDURE:

Steps to Create and implement pipelines in Azure:

1. Sign in to Azure DevOps and Navigate to Your Project
Log in to dev.azure.com, select your organization, and open the project where your Student Management System code resides.
2. Connect a Code Repository (Azure Repos or GitHub)
Ensure your application code is stored in a Git-based repository such as Azure Repos or GitHub. This will be the source for triggering builds and deployments in your pipeline.
3. Create a New Pipeline
Go to the Pipelines section on the left panel and click “Create Pipeline”.
Choose your source (e.g., Azure Repos Git or GitHub), and then select the repository containing your project code.
4. Choose the Pipeline Configuration
You can select either the YAML-based pipeline (recommended for version control and automation) or the Classic Editor for a GUI-based setup.
If using YAML, Azure DevOps will suggest a template or allow you to define your own.
5. Define Build Stage (CI - Continuous Integration) from YAML file
6. Install dependencies (e.g., npm install, dotnet restore)
7. Build the application (dotnet build, npm run build)
8. Run unit tests (dotnet test, npm test)
9. Publish build artifacts to be used in the release stage

10. Save and Run the Pipeline for the First Time

Save the YAML or build definition and click “Run”.

Azure will fetch the latest code and execute the defined build and test stages.

11. Configure Continuous Deployment (CD)

Navigate to the Releases tab under Pipelines and click “New Release Pipeline”.

Add an Artifact (from the build stage) and create a new Stage (e.g., Development, Production).

12. Configure the CD stage with deployment tasks such as deploying to Azure App Service, running database migrations or scripts, and restarting services using the Azure App Service Deploy task linked to your subscription and app details.

13. Set Triggers and Approvals

Enable continuous deployment trigger so the release pipeline runs automatically after a successful build.

For production environments, configure pre-deployment approvals to ensure manual verification before release.

14. Monitor Pipelines and Manage Logs

View all pipeline runs under the Runs section.

Check logs for build/test/deploy stages to debug any errors.

You can also integrate email alerts or Microsoft Teams notifications for build failures.

15. Review and Maintain Pipelines

Regularly update your pipeline tasks or YAML configurations as your application grows.

Ensure pipeline runs are clean and artifacts are stored securely.

Integrate quality gates and code coverage policies to maintain code quality.

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Library

Test Plans

Artifacts

Project settings

This run is being retained as one of 3 recent runs by pipeline.

View retention leases

Summary

Code Coverage

Individual CI by Barath-aiml

Repository and version

Barath-aiml/Software-Construction

main 734e4d22

Time started and elapsed

Just now

10s

Related

0 work items

0 artifacts

Tests and coverage

Get started

View change

Parameters

Stages

Jobs

Stage

1 job completed

10s

Job

10s

Run Stage

RESULT:

Thus the pipelines for the given project “E-commerce product uploader has been executed successfully

EXP NO: 10
DATE:

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the E-commerce product uploader project.

GitHub Project Structure

The screenshot displays a GitHub repository interface for 'Barath-aiml / Software-Construction'. The breadcrumb navigation shows the path: 'Software-Construction / ecommerce-platform / ecommerce /'. Below the navigation bar, a table lists the repository's files and folders. The table has three columns: 'Name', 'Last commit message', and 'Last o'. The listed items are: '..' (parent directory), 'css' (folder), 'db' (folder), 'js' (folder), 'README.md' (file), and 'index.html' (file). All files and folders show the commit message 'Add files via upload' and a commit hash 'bc2c6d2' from '1 hour ago'. Below the table, the 'README.md' file is open, showing the project title 'ShopFlow — E-Commerce Product Management Platform' with a shopping cart icon, followed by a section titled 'Project Structure' with a folder icon.

Name	Last commit message	Last o
..		
css	Add files via upload	1
db	Add files via upload	1
js	Add files via upload	1
README.md	Add files via upload	1
index.html	Add files via upload	1

README.md

ShopFlow — E-Commerce Product Management Platform

Project Structure

Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.

2116231501146

CS23432